# Use of biological resources

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

## 1 · Part 1

Paper 1 & 2 | Food production | [1]

**Answer: C**

### Worked solution

A control is the same set-up with one factor changed so you can see what causes the result. To show that living yeast causes the gas, keep everything the same but use dead (boiled and cooled) yeast. Cooling it first means the sugar solution is not heated by the yeast.

## 2 · Part 1

Paper 1 & 2 | Food production | [1]

**Answer: B**

### Worked solution

Respiration releases energy, and some is lost as heat. In a large fermenter this heat builds up. The water jacket keeps the temperature close to the optimum for the enzymes of the micro-organisms, so they grow and make product as fast as possible.

## 3 · Part 1

Paper 2 only | Food production | [1]

**Answer: D**

### Worked solution

"Inter" means between: interspecific predation is between different species. "Intra" means within: intraspecific predation is when fish eat smaller fish of their own species. Nets keep out predators of other species.

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## 4 · Part 1 Paper 1 & 2 Selective breeding [1]

**Answer: A**

**Explanation**

Breeding from only the highest-yielding animals for many generations reduces genetic variation, so if a new disease appears, few animals may have alleles for resistance and most could be affected. Selective breeding does not transfer genes from another species (B) and does not produce clones (C); the cattle still reproduce sexually. The high milk yield is inherited, which is why selective breeding works (D is wrong).

## 5 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [1]

**Answer: C**

**Explanation**

Transgenic means genetic material has been transferred from one species to a different species, so a bacterium with a human gene is transgenic. A plant from a runner (A) and a cloned sheep (B) are genetically identical to one parent and contain no genes from another species. Crossing two wheat varieties (D) is selective breeding within one species.

## 6 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [1]

**Answer: A**

**Explanation**

A vector carries DNA into another cell. Viruses naturally insert their genetic material into the cells they infect, so a useful gene added to a virus is carried into the host cell. Plasmids are found in bacteria, not viruses (B). Viruses do not make restriction enzymes; scientists add restriction enzymes to cut DNA (C). Viruses are not cells and do not reproduce by cell division (D).

## 7 · Part 1 Paper 2 only Cloning [1]

**Answer: D**

**Explanation**

In micropropagation, the cells of the explant divide by mitosis, which produces genetically identical cells, so all the new plants are genetically identical to the parent plant. Meiosis produces genetically different cells (A). Being sterile prevents infection by micro-organisms but does not prevent mutations (B). Micropropagation is asexual and involves no pollen or fertilisation (C).

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## 8 · Part 1 Paper 2 only Cloning [1]

**Answer: B**

**Explanation**

*In vitro* means grown outside a living organism, in a container such as a test tube or Petri dish in a laboratory, under sterile conditions. Soil in a glasshouse (A) and a field under a polythene tunnel (D) are where the plants may be grown later, not *in vitro* culture. Inside the parent plant (C) would be *in vivo*.

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## 9 · Part 1 Paper 1 & 2 Food production [8]

**Mark scheme**

**(a)** **[2]**

*Any two from:*

- protects the crop from frost OR cold OR wind OR rain OR bad weather (1)
- keeps out pests (1)
- artificial lighting can be used to increase light intensity OR to give longer periods of light (1)
- the water supply can be controlled (1)
- crops can be grown out of season OR the growing season is longer (1)

**IGNORE** references to carbon dioxide or temperature

**(b)** **[2]**

- (7.0 − 4.0) ÷ 4.0 × 100 (1)
- 75 % (1)

**ALLOW** 75 with no unit for 2 marks

**(c)** **[3]**

*Any three from:*

- carbon dioxide is a raw material OR reactant for photosynthesis (1)
- carbon dioxide concentration OR temperature was a limiting factor (1)
- higher temperature increases the rate of the enzyme-controlled reactions of photosynthesis OR molecules have more kinetic energy (1)
- so the rate of photosynthesis increases (1)
- more glucose OR sugar is made, which is used for growth OR to make fruit (1)

**(d)** **[1]**

- enzymes would be denatured, so the rate of photosynthesis would fall (1)

**ALLOW** heating to this temperature would cost more than the value of any extra yield

**ALLOW** plants would lose too much water OR wilt

**Worked solution**

(a) A glasshouse protects plants from bad weather and frost, can keep pests out, and lets the grower control light (artificial lights) and water. Crops can also be grown out of season.

(b) The increase is 7.0 − 4.0 = 3.0 kg per plant. Dividing by the original 4.0 gives 0.75; multiplying by 100 gives 75 %. Check: 4.0 × 1.75 = 7.0.

(c) Photosynthesis uses carbon dioxide and water to make glucose. In glasshouse 1 both carbon dioxide and temperature were limiting. More carbon dioxide gives more raw material, and a higher temperature speeds up the enzymes that control photosynthesis. More glucose is made, so the plants grow more and produce more tomatoes.

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(d) At very high temperatures enzymes denature, so photosynthesis slows down. Heating is also expensive, so the extra cost would not be worth it.

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## 10 · Part 1

Paper 1 & 2 Food production [7]

**Mark scheme**

**(a)** **[2]**

- as the mass of fertiliser increases, the yield increases (1)
- the yield levels off OR stops increasing above about 150–200 kg per hectare (1)

**ALLOW** correctly quoted data, e.g. yield stays at 6.5 tonnes per hectare from 200 kg per hectare

**(b)** **[2]**

- 6.4 − 3.0 = 3.4 tonnes per hectare (1)
- 3.4 × 12 = 40.8 tonnes (1)

**ALLOW** 41 tonnes

**ECF** their difference × 12

**(c)** **[2]**

- nitrate ions are used to make amino acids OR proteins (1)
- (proteins are needed) for growth OR to make new cells OR to make enzymes, so the plants grow larger (1)

**(d)** **[1]**

- there is little or no extra yield above 150 kg per hectare, so the extra fertiliser would waste money (1)

**ALLOW** extra fertiliser may be washed into rivers OR lakes and cause pollution

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**Worked solution**

(a) The yield rises from 3.0 to 6.4 tonnes per hectare as fertiliser increases to 150 kg per hectare, then only reaches 6.5 and stays there. Above about 150 kg per hectare, something other than nitrate is limiting growth.

(b) Extra yield per hectare = 6.4 − 3.0 = 3.4 tonnes. For 12 hectares: 3.4 × 12 = 40.8 tonnes. Check: 6.4 × 12 = 76.8 and 3.0 × 12 = 36.0; 76.8 − 36.0 = 40.8.

(c) Plants absorb nitrate ions from the soil and use them to make amino acids and proteins. Proteins are needed to make new cells and enzymes, so the plants grow more and give a higher yield.

(d) Going from 150 to 250 kg per hectare adds almost no yield, so the farmer would pay for fertiliser that gives no benefit.

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## 11 · Part 1 Paper 1 & 2 Food production [7]

**Mark scheme**

**(a)** **[1]**

- pests eat OR damage the crop, reducing the yield OR quality (1)

**ALLOW** pests can spread disease to the crop

**(b)** **[2]**

- the pesticide reduced the number of whitefly faster OR more at first (from 40 to 4 by week 2) (1)
- but numbers then increased again (to 41 by week 10), while the wasp reduced numbers more slowly but kept them low (5 at week 10) (1)

**(c)** **[4]**

*Indicative points, 1 mark each, max 4. For 4 marks the answer must include at least one advantage and at least one disadvantage.*

*Advantages:*

- the control organism reproduces, so the control lasts a long time OR does not need to be reapplied (1)
- it is specific, so it does not harm other organisms (1)
- no chemical residues on the crop OR in the environment (1)
- pests do not become resistant (1)

*Disadvantages:*

- it acts more slowly than a pesticide (1)
- it does not kill all the pests OR some pests remain (1)
- the control organism may feed on other species OR may itself become a pest (1)
- the control organism may die out or leave if pest numbers fall (1)

**ALLOW** pesticides kill pests quickly as a disadvantage of biological control

**Worked solution**

(a) Pests feed on crop plants and damage them, so the yield falls.

(b) In P the pesticide killed most whitefly within 2 weeks (40 to 4), but the survivors reproduced and by week 10 numbers were back to 41. In Q the wasp worked more slowly, but numbers kept falling to 5 because the wasps keep reproducing and killing whitefly.

(c) Biological control uses a living organism, such as the wasp, that preys on or parasitises the pest. Because it reproduces, it keeps working without being reapplied; it targets the pest, leaves no chemicals on food, and pests do not become resistant to it. However, it is slow, it never removes every pest, and the control organism could attack other species or become a pest itself.

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## 12 · Part 1 Paper 1 & 2 Food production [8]

**Mark scheme**

**(a)** **[2]**

- glucose on the left of the arrow (1)
- ethanol + carbon dioxide on the right of the arrow (1)

**ALLOW** alcohol for ethanol

**ALLOW** sugar for glucose

**REJECT** oxygen as a reactant, water as a product

**(b)** **[2]**

- carbon dioxide gas is produced (1)
- the gas forms bubbles that are trapped in the dough, so the dough expands (1)

**(c)** **[2]**

- enzymes in the yeast work faster at warm temperatures OR closer to their optimum temperature (1)
- so respiration is faster and more carbon dioxide is produced (so the dough rises faster) (1)

**(d)** **[2]**

- the high temperature kills the yeast OR denatures the yeast's enzymes (1)
- so respiration stops and no more carbon dioxide is produced (1)

**IGNORE** ethanol evaporates

**Worked solution**

(a) glucose → ethanol + carbon dioxide. No oxygen is used.

(b) The carbon dioxide forms bubbles that are trapped in the dough, so it rises.

(c) Respiration is controlled by enzymes. In a warm place the enzymes are nearer their optimum temperature, so the yeast respires faster and makes more carbon dioxide.

(d) At 220 °C the enzymes are denatured and the yeast dies. Respiration stops, so no more gas is made.

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## 13 · Part 1 Paper 1 & 2 Food production [9]

**Mark scheme**

**(a)** **[2]**

- the oil stops oxygen (from the air) reaching the mixture (1)
- so the yeast respires anaerobically (1)

**(b)** **[1]**

- 10.1 $cm^3$ (1)

**(c)** **[2]**

- 15.7 ÷ 10 (1)
- 1.57 $cm^3$ per minute (1)

**ALLOW** 1.6 $cm^3$ per minute

**(d)** **[1]**

- the volume of gas OR carbon dioxide collected (in 10 minutes) (1)

**(e)** **[1]**

- concentration of glucose OR mass OR concentration of yeast OR type of yeast OR thickness of oil layer OR pH (1)

**REJECT** temperature, time, volume of solution

**(f)** **[2]**

- at 50 °C the enzymes (in the yeast) are denatured OR the active site changes shape (1)
- so the rate of respiration is lower and less carbon dioxide is produced (1)

**Worked solution**

(a) Oil floats on the mixture and stops oxygen dissolving in, so the yeast must respire anaerobically, the process being investigated.

(b) (9.8 + 10.4 + 10.1) ÷ 3 = 30.3 ÷ 3 = 10.1 $cm^3$.

(c) Rate = 15.7 ÷ 10 = 1.57 $cm^3$ per minute. Check: 1.57 × 10 = 15.7.

(d) The dependent variable is what is measured: the volume of gas collected.

(e) Anything else that affects respiration rate must stay the same, for example the glucose concentration or the amount of yeast.

(f) Enzymes work faster up to their optimum, so more gas is made at 40 °C. At 50 °C some enzymes are denatured: the active site changes shape and the substrate no longer fits, so respiration slows and less gas is made.

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## 14 · Part 1 Paper 1 & 2 Food production [7]

**Mark scheme**

**(a)** **[2]**

- to kill bacteria OR micro-organisms already in the milk (1)
- which could compete with the *Lactobacillus* OR spoil the yoghurt OR cause disease (1)

**ALLOW** to sterilise OR pasteurise the milk for 1 mark

**(b)** **[2]**

- adding the bacteria to hotter milk would kill them OR denature their enzymes (1)
- 42 °C is (close to) the optimum temperature for the bacteria (to grow and respire) (1)

**(c)** **[3]**

*Any three from:*

- the bacteria grow OR reproduce (1)
- they use lactose OR sugar in respiration OR they ferment lactose (1)
- they produce lactic acid (1)
- the pH falls OR the milk becomes acidic (1)
- the milk thickens OR the milk proteins clot OR coagulate (1)
- the yoghurt gets a sour taste (1)

**Worked solution**

(a) Heating kills the micro-organisms already in the milk so they cannot compete with *Lactobacillus*, spoil the yoghurt or make people ill.

(b) At 85 °C the bacteria would be killed. At 42 °C, close to their optimum, they grow and respire quickly.

(c) *Lactobacillus* bacteria multiply and use the lactose in the milk, producing lactic acid. The milk becomes acidic (pH falls), which makes the milk proteins thicken and gives yoghurt its sour taste.

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## 15 · Part 1 Paper 1 & 2 Food production [8]

**Mark scheme**

**(a)** **[2]**

- air provides oxygen for aerobic respiration (so the fungus grows) (1)
- the air is sterile so that no other micro-organisms enter OR to prevent contamination (1)

**(b)** **[2]**

*Any two from:*

- mixes the fungus with the nutrients (1)
- mixes oxygen through the culture (1)
- keeps the temperature even throughout (1)
- stops the fungus settling at the bottom (1)

**(c)** **[2]**

- glucose is used in respiration to release energy (1)
- nitrogen is needed to make amino acids OR proteins (for growth) (1)

**(d)** **[2]**

- enzymes in the fungus work best at the optimum pH (1)
- a change in pH changes the shape of the active site OR denatures enzymes, so growth OR product formation slows (1)

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**Worked solution**

(a) The fungus respires aerobically, so it needs oxygen from the air. The air is sterile so no other micro-organisms enter that could compete with the fungus or contaminate the food.

(b) Stirring (agitation) mixes nutrients, oxygen and fungus, and spreads heat evenly so there are no hot spots.

(c) Glucose is the fuel for respiration, releasing energy for growth. Nitrogen is needed to make amino acids and proteins, which the fungus needs to grow (and which make the food protein-rich).

(d) The fungus's enzymes have an optimum pH. If the pH moves away from it, the active site changes shape, enzymes work more slowly or are denatured, and growth slows.

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## 16 · Part 1 Paper 2 only Food production [9]

**Mark scheme**

**(a)** **[2]**

- as stocking density increases, the mean increase in mass (growth) decreases (1)
- as stocking density increases, the percentage of fish with disease increases (1)

**ALLOW** correctly quoted data, e.g. from 820 g to 430 g, or from 2 % to 31 %

**(b)** **[2]**

*Any two from:*

- fish are closer together, so pathogens spread more easily between fish (1)
- more waste in the water lowers water quality (1)
- less oxygen per fish OR fish are stressed, so they are less able to resist disease (1)
- fish are more likely to injure each other, giving entry points for pathogens (1)

**(c)** **[2]**

*Any two from:*

- waste and uneaten food are decomposed by bacteria (1)
- the bacteria use up oxygen in the water (for respiration), so less oxygen is available for the fish (1)
- waste products can be toxic to fish (1)
- waste encourages the growth of pathogens OR increases disease (1)

**(d)** **[3]**

*Any three from:*

- select the fastest-growing fish as parents (1)
- breed them together (1)
- select the fastest-growing offspring and breed them (1)
- repeat for many generations (1)

**Worked solution**

(a) At 10 fish per $m^3$ each fish gained 820 g and 2 % had disease; at 80 fish per $m^3$ each gained only 430 g and 31 % had disease. More crowding means slower growth and more disease.

(b) Crowded fish touch more often, so pathogens pass from fish to fish more easily. More fish make more waste, and each fish gets less oxygen, which stresses the fish.

(c) Bacteria decompose waste and uneaten food, using oxygen as they respire, so the fish get less oxygen. Some waste, such as ammonia, is also toxic, and dirty water helps disease spread.

(d) Choose the fastest-growing fish and breed them. From their offspring, choose the fastest-growing again and breed those. Repeating over many generations gives fish that grow faster.

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## 17 · Part 1 Paper 1 & 2 Selective breeding [6]

**Mark scheme**

**(a)** **[3]**

*Any three from:*

- cross variety P with variety Q OR transfer pollen from one variety to the other (1)
- grow the offspring and select plants with high yield and resistance to rust (1)
- cross OR breed these selected plants together (1)
- repeat over many generations (1)

**(b)** **[2]**

- not all offspring inherit both characteristics OR the alleles for both characteristics (1)
- repeated selection is needed until all the offspring have both characteristics (1)

**ALLOW** until the new variety breeds true

**(c)** **[1]**

- to find out which plants are resistant (the plants that are not infected) (1)

**Worked solution**

(a) Cross P with Q by transferring pollen. Grow the seeds, and from the offspring choose the plants that are high-yielding and resistant to rust. Cross these chosen plants and repeat the selection for many generations.

(b) Offspring inherit a mixture of alleles, so after one cross only some plants have both features, and their offspring may not all have both. Selection must be repeated until all the offspring have both characteristics.

(c) Resistance only shows when the fungus is present. Plants that stay healthy are resistant and can be chosen; infected plants are rejected.

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## 18 · Part 1 Paper 1 & 2 Selective breeding [5]

**Mark scheme**

**(a)** **[2]**

- 231 − 180 = 51 (1)
- 51 ÷ 4 = 12.75 eggs (per hen per year per generation) (1)

**ALLOW** 12.8 OR 13

**ALLOW** 1 mark for 51 ÷ 5 = 10.2

**(b)** **[2]**

- each round of selection reduces genetic variation OR most hens now have the alleles for high egg production (1)
- so there is less variation to select from OR egg production is approaching the maximum a hen can achieve (1)

**(c)** **[1]**

- choose cockerels whose mothers OR sisters OR female relatives laid the most eggs (1)

**Worked solution**

(a) The total increase is 231 − 180 = 51. From generation 1 to generation 5 there are 4 steps, not 5, so the mean increase per generation is 51 ÷ 4 = 12.75. Check: 180 + 4 × 12.75 = 231.

(b) The increases were 16, 14, 12 and 9. Each round of selection breeds from fewer types of hen, so genetic variation falls. Once most hens carry the alleles for high egg laying, selection achieves less, and hens are also nearing a biological limit.

(c) A cockerel's own egg laying cannot be measured, but he carries alleles from his parents. Choosing cockerels from hens that laid many eggs makes it likely they pass on the alleles for high egg production.

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## 19 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [7]

**Mark scheme**

**(a)** **[1]**

- restriction enzyme (1)

**(b)** **[2]**

- restriction enzymes cut DNA at specific sites OR specific base sequences (1)
- so the gene and the opened plasmid have matching OR complementary (sticky) ends that can join (1)

**(c)** **[1]**

- joins the gene into the plasmid OR joins pieces of DNA together (1)

**(d)** **[1]**

- recombinant DNA (1)

**(e)** **[2]**

- the plasmid with the human gene is inside the bacterium (1)
- the bacterium uses the gene (base sequence) to make the protein OR human growth hormone (1)

**Worked solution**

(a) Restriction enzymes cut DNA.

(b) Each restriction enzyme cuts at a specific base sequence. Using the same enzyme gives the human gene and the opened plasmid matching ends, so they fit together.

(c) Ligase joins the gene into the plasmid.

(d) DNA made from two different organisms is called recombinant DNA.

(e) A gene is a section of DNA that codes for a protein. The bacterium reads the human gene in its plasmid and makes human growth hormone.

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## 20 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [7]

**Mark scheme**

**(a)** **[3]**

- n = 240 ÷ 30 = 8 (1)
- $4.0 \times 10^3 \times 2^8$ OR $4.0 \times 10^3 \times 256$ (1)
- $1.0 \times 10^6$ OR $1.02 \times 10^6$ OR $1.024 \times 10^6$ (1)

**ALLOW** 1 024 000 for 2 marks only (not in standard form)

**ECF** their n

**(b)** **[3]**

*Any three from:*

- bacteria reproduce rapidly OR divide very often (1)
- each new bacterium has a copy of the plasmid OR insulin gene (1)
- the fermenter provides optimum conditions (e.g. nutrients, temperature, pH, oxygen) for fast growth (1)
- every bacterium makes insulin, which is collected OR extracted and purified (1)

**(c)** **[1]**

- it is human insulin, so it is less likely to cause an allergic reaction OR side effects (1)

**ALLOW** large amounts can be made OR supply does not depend on pigs

**ALLOW** acceptable to people who do not use pig products

**Worked solution**

(a) In 4 hours = 240 minutes, the bacteria divide 240 ÷ 30 = 8 times. Number = $4.0 \times 10^3 \times 2^8 = 4000 \times 256 = 1\,024\,000 = 1.0 \times 10^6$ (2 s.f.). Check: 1 024 000 halved 8 times gives 4000.

(b) Bacteria divide very quickly, and each copy keeps the plasmid with the insulin gene, so the number of insulin-making cells grows fast. The fermenter keeps nutrients, temperature, pH and oxygen at the best levels, and the insulin is extracted and purified.

(c) GM bacteria make human insulin, which is less likely to cause a reaction than pig insulin, and supply does not depend on pigs.

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## 21 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [8]

**Mark scheme**

**(a)** **[2]**

- (0.4 − 2.5) ÷ 2.5 × 100 (1)
- (decrease of) 84 % (1)

**ALLOW** −84 %

**(b)** **[2]**

- fewer caterpillars feed on the leaves OR less leaf damage (1)
- so more leaf area OR more photosynthesis, so more growth OR more yield (1)

**(c)** **[4]**

*Indicative points, 1 mark each, max 4. For 4 marks the answer must include at least one advantage and at least one disadvantage.*

*Advantages:*

- higher yield (15 % more) OR more food from the same area (1)
- less insecticide is used, so fewer other insects OR organisms are harmed (1)
- lower cost of insecticide OR less spraying (1)

*Disadvantages:*

- caterpillars may become resistant to the toxin (1)
- the toxin may harm other insects that are not pests, e.g. pollinators (1)
- the gene may spread to wild plants by pollen (1)
- GM seed may be expensive OR farmers must buy new seed (1)
- some people are concerned about eating GM food OR long-term effects are unknown (1)

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**Worked solution**

(a) Change = 0.4 − 2.5 = −2.1 kg per hectare per year. −2.1 ÷ 2.5 × 100 = −84 %, a decrease of 84 %. Check: 2.5 × 0.16 = 0.4.

(b) The toxin kills caterpillars that eat the leaves. With more leaf area left, the plant photosynthesises more, makes more glucose and grows more, giving a higher yield.

(c) For: more food per hectare, and far less insecticide, which saves money and harms fewer other organisms. Against: caterpillars may evolve resistance, the toxin may harm harmless insects, the gene could spread to wild plants, the seed costs more, and some people worry about eating GM food.

## 22 · Part 1 Paper 1 & 2 Genetic modification (genetic engineering) [6]

**Mark scheme**

**(a)** **[3]**

*Any three from:*

- weeds compete with the soya for light OR water OR mineral ions (1)
- without weeds, the soya gets more light, so more photosynthesis (1)
- more water OR mineral ions are available to the soya (1)
- so the soya grows more OR gives a higher yield (1)

**(b)** **[2]**

*Any two from:*

- the herbicide-resistance gene could spread to weeds (e.g. by pollen), producing resistant weeds (1)
- farmers may use more herbicide, which could harm other plants OR organisms (1)
- loss of wild plants reduces food OR habitats for other organisms OR reduces biodiversity (1)
- GM seed may be expensive (1)
- some people are concerned about eating GM food (1)

**(c)** **[1]**

- the gene is part of the plant's DNA, so it is passed on (inherited) in reproduction (1)

**Worked solution**

(a) Weeds take light, water and mineral ions the crop needs. With weeds killed, the soya gets more light for photosynthesis and more water and ions, so it grows more.

(b) The gene might reach related weeds through pollen, making herbicide-resistant weeds. Spraying more herbicide could also kill wild plants and reduce food and habitats for other organisms.

(c) The inserted gene is now part of the soya's own DNA, so it is copied into the seeds and inherited by the next generation.

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## 23 · Part 1 Paper 2 only Cloning [8]

**Mark scheme**

**(a)** **[4]**

*Any four from:*

- small pieces (explants) are cut from the parent plant, e.g. from the shoot tip (1)
- the explants are sterilised OR aseptic conditions are used (1)
- explants are placed on a growth medium (agar) containing nutrients (1)
- the medium contains plant hormones OR auxin (to stimulate growth) (1)
- the cells divide by mitosis to form many small shoots OR plantlets (1)
- the plantlets are separated and can be cultured again (1)
- when roots have grown, the plantlets are moved to compost OR soil (1)

**(b)** **[2]**

- $n = 20 \div 5 = 4$ (1)
- $6^4 = 1296$ plantlets (1)

**ECF** their n

**(c)** **[2]**

- there is no genetic variation (1)
- so if a disease OR pest affects one plant, it could affect all of them OR the whole crop could be lost (1)

**ALLOW** all would be affected by the same change in conditions

**Worked solution**

(a) Tiny explants are cut from a parent plant, often from the shoot tip, and sterilised. They are grown *in vitro* on sterile agar with nutrients and plant hormones. The cells divide by mitosis to form shoots, which can be split and grown again. Plantlets with roots are moved to compost.

(b) 20 weeks is 4 cycles. $6^4 = 6 \times 6 \times 6 \times 6 = 1296$. Check: $6^2 = 36$ and $36 \times 36 = 1296$.

(c) All the orchids have the same genes, so if one is not resistant to a disease, none are. One disease could destroy the whole stock.

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## 24 · Part 1 Paper 2 only Cloning [7]

**Mark scheme**

**(a)** **[4]**

*Any four from:*

- the nucleus is removed from an egg cell (of sheep Y) to give an enucleated egg cell (1)
- a diploid nucleus is taken from a mature udder cell (of sheep X) (1)
- the diploid nucleus is inserted into the enucleated egg cell (1)
- the cell is stimulated to divide, e.g. by an electric shock (1)
- the cell divides by mitosis to form an embryo (1)
- the embryo is placed in the uterus of the surrogate mother (sheep Z) (1)

**(b)** **[2]**

- Dolly's genes OR DNA OR nucleus came from the udder cell of sheep X (1)
- the nucleus of the egg cell from sheep Y was removed, and the surrogate (sheep Z) only provided the uterus OR passed on no genes (1)

**ALLOW** Dolly is genetically identical to sheep X OR is a clone of sheep X

**IGNORE** mitochondrial DNA

**(c)** **[1]**

- it has two sets of chromosomes OR two copies of each chromosome (1)

**ALLOW** it has the full number of chromosomes

**Worked solution**

(a) The nucleus was removed from an egg cell of sheep Y to make an enucleated egg cell. A diploid nucleus from an udder cell of sheep X was put in, and an electric shock made the cell divide by mitosis into an embryo. The embryo was placed in the uterus of the surrogate, sheep Z.

(b) Genes are in the nucleus. Dolly's nucleus came from sheep X and the egg's own nucleus had been removed, so all her DNA came from sheep X. The surrogate only carried the embryo and supplied no genes.

(c) Diploid means two sets of chromosomes, like a normal body cell.

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## 25 · Part 1 Paper 2 only Cloning [7]

**Mark scheme**

**(a)** **[2]**

- in sexual reproduction, only some of the offspring would inherit the human gene (1)
- clones are genetically identical, so all of them have the gene OR make the protein (1)

**ALLOW** about half the offspring would inherit the gene

**(b)** **[3]**

- 2.0 × 3.0 = 6.0 g per goat per day (1)
- 6.0 × 300 = 1800 g OR 1.8 kg per goat per year (1)
- 9.0 ÷ 1.8 = 5 goats (1)

**ECF** at each step

**ALLOW** correct alternative order of working

**(c)** **[1]**

- the protein can be collected without harming the goats (1)

**ALLOW** the goats can be milked many times, so the protein is produced continuously

**ALLOW** milk is easy to collect in large amounts

**(d)** **[1]**

- the nucleus came from a female cell, so all the clones have XX sex chromosomes (1)

**ALLOW** they are genetically identical to the female goat

**Worked solution**

(a) Breeding a transgenic goat with a normal goat mixes the alleles, so only some kids would get the human gene. Clones are genetically identical to the cell that provided the nucleus, so every cloned goat has the gene and makes the protein.

(b) One goat makes 2.0 × 3.0 = 6.0 g per day, so 6.0 × 300 = 1800 g = 1.8 kg per year. Goats needed = 9.0 ÷ 1.8 = 5. Check: 5 × 1.8 = 9.0 kg.

(c) Goats can be milked every day without harm, so collecting the protein is easy and it is made continuously.

(d) The nucleus came from a female goat, so its sex chromosomes are XX, and every clone is female too.

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